

100 Days of Azure DevOps

Day 66 Handout — Definitions assign effects (Deny, Audit, ...) at a scope — Blueprints are retired, initiatives are bundles

7 - Security, Compliance & Governance · Personal lab only · Educational content · Not a sales pitch

Architecture A — definition, initiative, assignment, effect

Object	What it is	Lab move
Definition	JSON: if/then. Built-in 'Require a tag on resource groups'	Use a built-in. Do not author custom JSON on day one
Initiative (policy set)	Bundle of definitions, one assignment	Literacy. Assign one definition today
Assignment	Definition/initiative + scope (MG / sub / RG) + parameters + effect overlay sometimes	Assign at lab RG or personal sub — not a work MG
Compliance blade	Compliant / non-compliant / not started	Create a violating RG and watch the state — then delete it
Blueprints	Retired. Do not learn a dead product to sound enterprise	Policy + landing-zone ideas are the current sentence

Architecture B — pick an effect like a culture choice, not a default checkbox

Effect	What Azure does	Use when
Audit	Resource exists; marked non-compliant	You need visibility first. Factories break if you Deny on day one
Deny	ARM PUT is blocked. Sin cannot land	Lab tag requirement. Production after Audit noise is understood
AuditIfNotExists	Related resource missing (e.g. diagnostics) → non-compliant	You are surveying a gap, not deploying it yet
DeployIfNotExists	Repair by deploying the missing related resource	You accept the managed identity Policy needs to deploy
Modify / Append / Disabled	Mutate fields, or park the assignment	Append a tag; Disabled = assignment exists but does nothing

Architecture C — pitfalls that never fire or block the wrong RG

Pitfall	What you see	Fix
Wrong scope	Compliance always green; you created the RG elsewhere	Assignment scope must ancestor the resource
Audit vs Deny confusion	'Policy is on' but untagged RG still created	Read the effect. Audit does not block
Not-started forever	Evaluation lag or wrong resource type	Trigger a scan; confirm the definition applies to resource groups
Custom JSON first	Broken if/then, empty compliance	Built-in require-tag-on-RGs, parameter tagName=env
Work management group	Employer policy in a personal series screenshot	Personal sub/RG only. Delete the test RG after

One-liner to remember

Policy is the grown-up saying 'no untagged RGs' so Finance does not hunt you with spreadsheets. Audit sees sin; Deny stops it; the effect is a choice.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal subscription or rg-day66. Built-in policy only. Clean up the violating RG so compliance is not a junk drawer.

- 1 Portal → Policy → Definitions. Search Require a tag on resource groups. Open it and read the effect (typically Deny) and the parameter tagName.
- 2 Policy → Assignments → Assign policy. Scope = personal sub or rg-day66. Parameter tagName=env. Note whether you leave Deny or set Audit for the experiment.
- 3 Try az group create -n rg-day66-nopolicy -l eastus with no env tag. If Deny: create fails. If Audit: create succeeds and Compliance shows non-compliant.
- 4 Create a compliant RG with --tags env=lab. Open Compliance blade. Screenshot states without any employer MG names.
- 5 Delete the non-compliant/junk RG. Do not leave Policy nagging a leftover forever. Optional CLI: az policy assignment create with the built-in definition ID.
- 6 Write docs/policy-day66.md: effect you used, what you observed, initiative vs definition in one line, Blueprints = retired.

Starter CLI — assign a built-in, then prove the effect

```
# Built-in: Require a tag on resource groups (effect: Deny)
# Definition ID 96670d01-0a4d-4649-9c89-2d3efa4e1136
SUB=$(az account show --query id -o tsv)
az policy assignment create --name require-rg-tag-lab \
  --display-name "Require env tag on RGs (lab)" \
  --scope /subscriptions/$SUB \
  --policy 96670d01-0a4d-4649-9c89-2d3efa4e1136 \
  --params '{"tagName":{"value":"env"}}'
# Prove Deny:
# az group create -n rg-day66-nopolicy -l eastus           # should fail
az group create -n rg-day66 -l eastus --tags env=lab      # should succeed
```

Done checklist

- Can name Deny vs Audit vs AuditIfNotExists vs DeployIfNotExists
- One built-in assignment visible on the compliance blade
- Proved the effect with a tagged vs untagged RG
- Junk RG deleted; Blueprints not in the notes as current

Tomorrow — Day 67

Compliance scanning in pipelines — SARIF as a format, not a product tour